

Project Summary & Problem Framing (10% of final project grade)

The purpose of this project was to see how well we could predict G3 grades based on various pieces of data pertaining to students. The objective was to determine when students would benefit from interventions or additional assistance. An additional challenge was to see how well we could predict G3 grades *without* using the grades from G1 or G2. It would be important to have the ability to predict G3 outcomes without including G1 and G2 grades because we could offer intervention to students sooner.

The key finding of this project shows that it is near impossible to predict G3 grades without including G2 and G1 grades. Including the G2 and G1 grades allowed our model to predict G3 grades, typically within about 1 point of the real grade. In contrast, without including the earlier grades, the prediction is off by an average of 2.5 points from the final grade.

My recommendation is that we use the model that includes G1 and G2 grades to predict final outcomes. This would allow us to intervene at the earliest possible time while also giving us the most accurate results in determining who will need interventions or assistance.

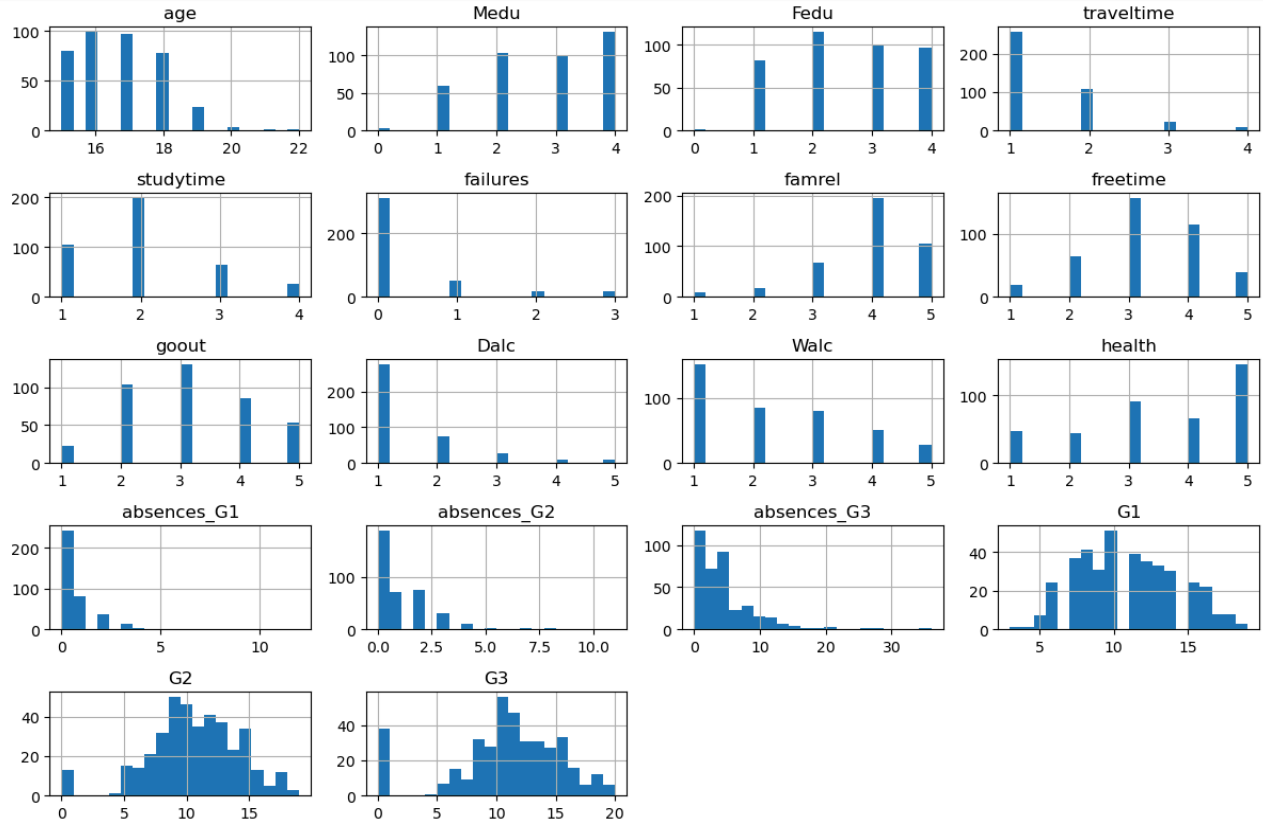
Data Overview (10% of final project grade)

The dataset includes information from 395 students attending two Portuguese schools. The data was originally collected from school reports and questionnaires. There were 35 total categories ranging from basic information like sex and age, to more complex information such as parent's education, studytime, and goals for higher education. Additionally, important categories such as failures and absences are included.

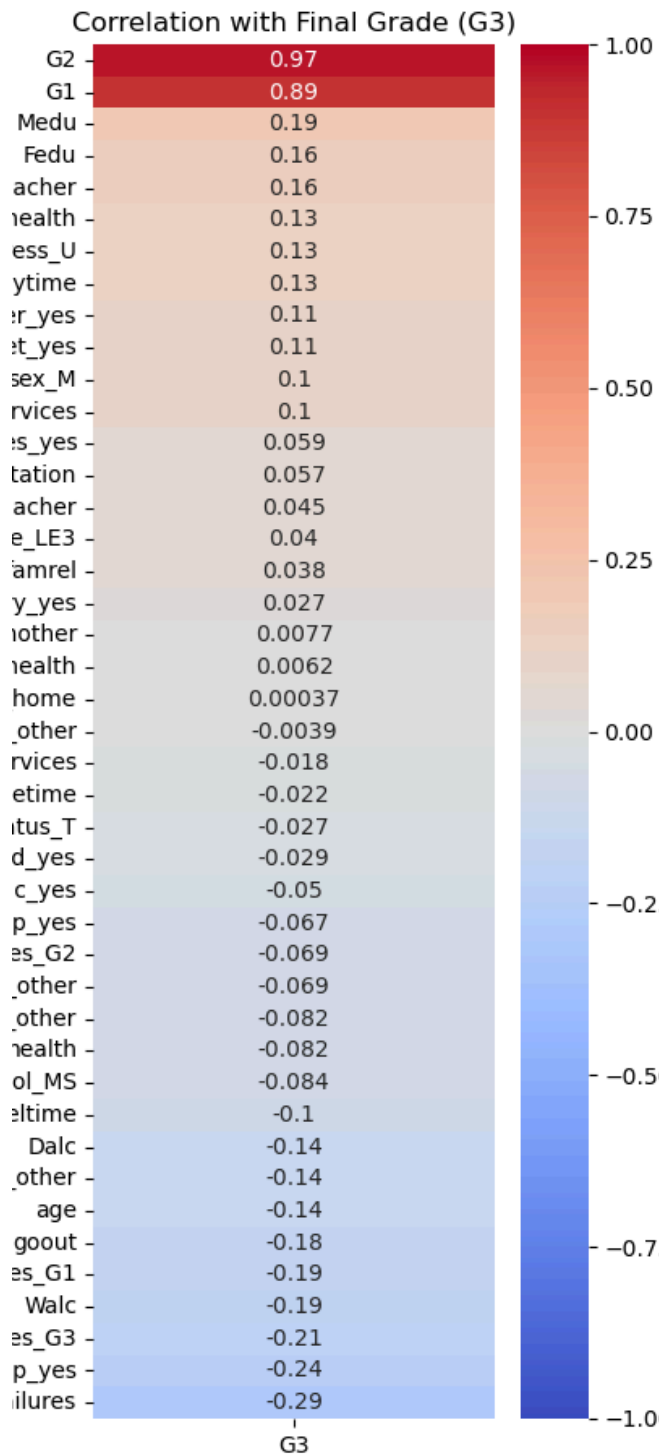
It should be noted that there were some missing values in the age and absences columns. I filled the missing data points with the average value of their respective category. Additionally, there were several scores of "0" for G3 and G2 that appeared to be outliers so I have excluded them from the final model.

I created a model with and without G1 and G2 grades to see how accurately we could predict G3 grades given the different information. The earlier we can determine which students need interventions or assistance, the better, as this can help us catch any students who might have a better chance of succeeding. Having a model that can predict G3 grades without any input for G1 or G2 would greatly influence how early these students can begin receiving assistance.

Analytical Insight (15% of final project grade)



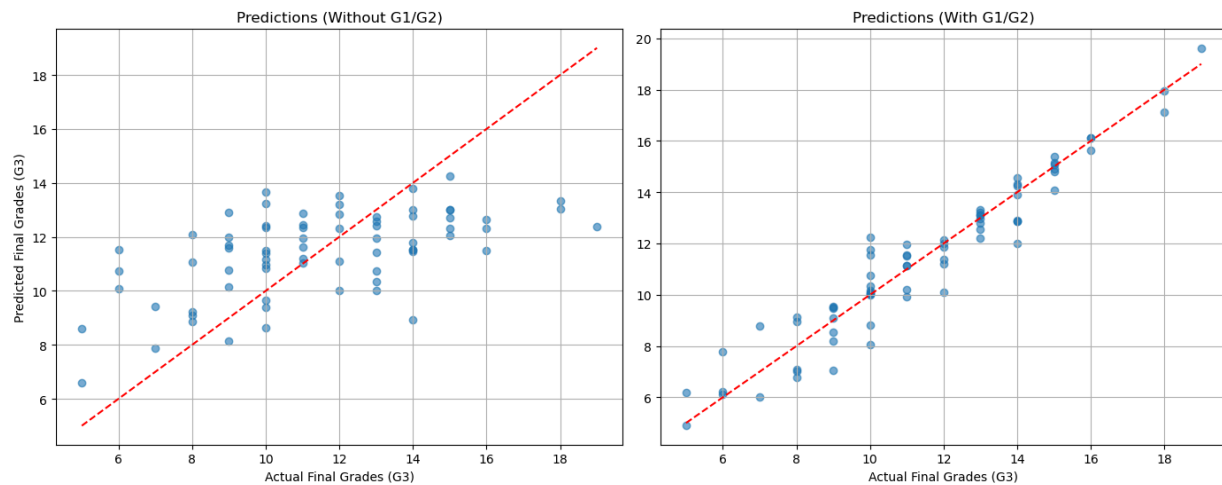
These histograms show the distributions for each of the numeric and ordinal categories. I used these histograms to look for any outliers in the data that might be useful to exclude. The two instances that showed the greatest improvement in my model after excluding them were the scores of "0" in both the G2 and G3 categories.



I used this heatmap to decide which categories to include in the model. This heatmap shows which categories are most strongly correlated with G3 grades. As you can see, G2 and G1 grades are very strongly correlated whereas every other category has a very weak correlation. After some testing and tweaks, I decided to use all of the categories with a correlation coefficient whose absolute value was greater than or equal to .13.

Key finding: G2 grades are the greatest predictor for G3 grades.

Comparison of Predictions



These charts show how well each of the models predicted final grades. As you can see, the data more closely fit the predictions when we include G1/G2 grades versus the model when G1/G2 grades are excluded. The red dashed line would represent a “perfect prediction.” As you can see, there are a lot of gaps between the red line and the data points on the left graph whereas the right graph has much less white space between the points and the “perfect prediction” line.

Throughout the project, I tested several different features to try to find the option which yielded the highest R^2 and lowest RMSE value for all of the models I tested (linear, SVM, and Lasso). In the end, the features that yielded the best results for me were the following: absences, failures, schoolsup, activities, higher, paid, nursery, Mjob, guardian, school, address, traveltime, health, Fedu, Walc, goout, studytime, and Medu.

Methodology (5% of final project grade)

I used three models to help predict the G3 grades: 1) linear regression 2) SVM regression, and 3) Lasso regression.

- 1) Linear regression is simply trying to draw a straight line through the data that leaves the smallest error possible. It assumes that while one category increases/decreases the output will also either increase/decrease.
- 2) SVM regression is also trying to find the best line to fit the data but it doesn't have to be a straight line. This line can be curved to better fit the data and further reduce error.
- 3) Lasso regression is very similar to linear regression but it tries to simplify the model by removing any unnecessary features, or the features that don't actually affect the final grade.

The metrics I used to determine if the strength of my models were the RMSE and R^2 . RMSE is simply how far the predictions are away from the actual values. The lower an RMSE is, the better. R^2 tells us how much of the real-world results can be explained using our model. A perfect score of 1.0 would tell us that we can perfectly explain the results. The lower the R^2 is, the worse our model is at predicting the real world results.

Key Results (5% of final project grade)

1. Present the results and metrics for all models, both with and without the G1/G2 features, with a clear designation of your best result.
2. Do you recommend implementing this model into production? Why or why not?
3. Provide actionable business recommendations based on your machine learning model and data analysis findings. Along with making recommendations about your model, include other recommendations derived from your analysis insights to guide Decision-making.

Model	RMSE (with grades)	R^2 (with grades)	RMSE (without grades)	R^2 (without grades)
Linear Regression	0.876	0.927	3.116	0.073
SVM Regression	1.130	0.878	3.089	0.092
Lasso Regression	1.314	0.836	3.260	-0.010
Tuned SVM Regression	0.879	0.920	2.554	0.323

Based on the final results of all of the models, the "Tuned" SVM regression provided the best results overall. However the Linear regression showed the best results when using the G1 and G2 grades. I do not recommend using *any* of the models without grades as none of them show a strong enough correlation to be of any use.

I recommend the use of the linear regression model as long as we use G1 and G2 grades when making predictions. The linear Model has a low RMSE and shows predictions are typically within 1 point of their actual G3 grade. Additionally, the R^2 shows that we are able to capture about 93% of the reasons why a student's final grades might vary. Or, put simply, it very closely follows the true outcomes.

Conclusion (5% of final project grade)

The entire project was a “success” in the sense that I was able to complete it. I found it quite fun to go back and look at all of the assignments for this course and see which pieces I could use and experiment with which parts would even be useful for this project.

In contrast to my successes, I found myself spending a lot of time making adjustments to improve the end result. I dedicated a lot of additional time to testing different combinations of features to see what I could do to improve the regressions without using the G1 and G2 grades. I assumed using a heatmap to chart correlations would give me a good idea right away of which features would be most beneficial to use but I was wrong because all of them had such weak correlations to G3 grades. My biggest downfall was that I was so focused on finding the “best” model that I didn’t stop to think that the entire idea was that it would be impossible to create a good one if we didn’t use the G1 and G2 grades in our predictions.

The next steps would be to use the models we created and see how the predictions work over the next year. Adjustments would probably need to be made year over year based on how well grades are predicted. Additionally, it would be useful to track students who receive any of the interventions or assistance and see how their grades affect predictions. The new data from tracking interventions and assistance could be useful in training the next model.